ARUN KRISHNAN

Cloud Data Engineer
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Portfolio: arun-krishnan-1996.github.io/portfolio

SUMMARY

Cloud Data Engineer with 4+ years of experience architecting real-time data pipelines, scalable cloud infrastructure, and distributed processing systems, specialising in CDC streaming, Apache Spark, and enterprise Azure solutions. Track record of reducing data latency by 90%, processing 1M+ events daily, and maintaining 99.9% pipeline uptime in production. Active open-source contributor to Apache StarRocks, with a merged bug fix cherry-picked into a stable release branch. Committed to leveraging data-driven insights to improve data management strategies.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Data Quality & Testing:** Unit Testing, Great Expectations (GE)
- **Container & Orchestration:** Docker, Azure Kubernetes, CI/CD, Container Registry
- **Data Science & Machine Learning:** Pandas, NumPy, Scikit-Learn, TensorFlow, Matplotlib
- **Databases & Storage:** PostgreSQL, Elasticsearch, StarRocks DB, Delta Lake, Apache Iceberg, Nessie (Data Lakehouse Catalog), Dremio
- **Big Data & Frameworks:** Apache Spark (PySpark), Apache Flink, Hive, Kafka, Apache Airflow, Debezium
- **Cloud Platform & ETL Services:** ADLS Gen2, Azure Data Factory, Azure Event Hub, Azure Fabric, Synapse, Azure Databricks, AWS S3

EXPERIENCE

AEDGE AICC India Pvt. Ltd. (Armada.ai) — **June 2025 - Present**
L2 Data Engineer — *Trivandrum*

- Designed and implemented a real-time data pipeline from PostgreSQL to Azure Synapse, enabling 100% CDC using Debezium and Azure Event Hub.
- Developed Spark jobs on Databricks to process streaming CDC data and load it into Synapse Data Warehouse, reducing data latency by 90%.
- Deployed a production StarRocks OLAP cluster on AKS (3 FE / 5 BE nodes) with automated disaster recovery to Azure Blob Storage, achieving RTO <15 min, RPO <5 min, and 99.99% data durability.
- Delivered an ML forecasting pipeline on Databricks, posting daily predictions to a REST API, improving forecast delivery speed by 80%.
- Orchestrated end-to-end workflows using Airflow, increasing pipeline reliability by 95% with automated retries and alerting.
- Deployed Debezium and Airflow on AKS, improving infrastructure scalability and reducing manual maintenance by 70%.
- Collaborated cross-functionally with Backend, AI/ML, Power BI, and Frontend teams to deliver integrated data solutions that drive business insights and product functionality.

Turbolab Technologies Pvt. Ltd. (ScrapeHero) — **July 2022 - June 2025**
Associate Data Engineer — *Kochi*

- Architected scalable data infrastructures using PySpark, Dremio, and Iceberg to handle large volumes of data, achieving a scalability improvement of up to 40%.
- Designed and implemented efficient ETL processes, reducing data processing time by 25% and enhancing data quality by 20%.
- Developed and optimised data models and schemas using dimensional modelling for effective data storage and retrieval, improving data access speed by 30%.
- Collaborated with cross-functional teams to integrate machine learning models into the data pipeline, increasing predictive analytics accuracy by 25%.
- Implemented data governance policies and procedures to ensure data security, compliance, and privacy, achieving a 95% data accuracy rate.
- Implemented Great Expectations (GE) to improve data quality by 67%, utilising data profiling, validation, and automated monitoring to ensure adherence to essential quality metrics like completeness, uniqueness, and format accuracy.

OPEN SOURCE CONTRIBUTIONS

StarRocks Contributor – [Release Note]

github.com/StarRocks/starrocks

Merged PR #73509 — "[BugFix] Fix FE startup failure during ADLS2 shared-data disaster recovery caused by AZURE_PATH_KEY validation," cherry-picked to branch-4.1.

Diagnosed a cross-module constant visibility gap (AZURE_PATH_KEY missing from Cloud Configuration Constants.PARAM_NAMES) that aborted FE node startup during ADLS2 disaster recovery, and fixed it by relocating the constant into the shared module (8 files changed).

Result: FE nodes now start reliably after ADLS2 DR restoration, supporting sub-15-minute recovery times in production; merged with 2 approvals.

PROJECTS

StarRocks Cluster Deployment & Disaster Recovery — *Armada (Professional)*

- Deployed a production StarRocks OLAP cluster on AKS (3 FE, 5 BE nodes) with automated snapshot-based disaster recovery to Azure Blob Storage, cross-zone replication, and automated restore procedures; added Prometheus + Grafana monitoring. Achieved RTO <15 min, RPO <5 min, and 99.99% data durability.

Technologies: *StarRocks, Kubernetes, Helm, Azure Blob Storage, Prometheus, Grafana, Python, Shell Script*

Data Ingestion Pipeline: PostgreSQL to Synapse — *Armada (Professional)*

- Built a real-time ingestion pipeline capturing 100% of CDC changes from PostgreSQL via Debezium, streaming 1M+ events/day through Azure Event Hub, processed with Spark on Databricks, and loaded into Azure Synapse with a 90% reduction in data latency. Achieved 99.9% pipeline uptime with automated recovery and monitoring, ensuring reliable and timely data delivery for downstream analytics.

Technologies: *PostgreSQL, Debezium, Event Hub, Spark, Databricks, Delta Lake, Synapse, AKS*

ML Model Integration with POST API using Databricks — *Armada (Professional)*

- Integrated a time-series forecasting ML model in Databricks to generate daily predictions, posting results to a secured POST API, reducing manual reporting by 85%. Automated the end-to-end pipeline using Apache Airflow, improving model delivery speed and reliability by 90%.

Technologies: *Databricks, PySpark, MLflow, REST API, Apache Airflow, Kubernetes (AKS), Python*

E-commerce Data Pipeline — *Turbolab (Professional)*

- Designed a scalable pipeline architecture that consumes data from Kafka (source) using PySpark as RDD, improving processing speed by 45% compared to a standard DataFrame. Transforms and compacts the data into Iceberg and Postgres tables, validates data quality (67% improvement), then writes it into the Iceberg data lake and Postgres table (destination).

Technologies: *Python, PySpark, Iceberg, Nessie, Postgres, Shell Script, Great Expectations (GE)*

Product's Customer Reviews Data Batch Processing ELT Pipeline — *Turbolab (Professional)*

- Streamlined data ingestion from Kafka into a data lake, enforced data quality checks using GE in Airflow, and orchestrated a Spark batch pipeline to implement transformations and aggregations, ensuring clean data is delivered to a Postgres database with 100% assurance that the data successfully reached the Postgres DB.

Technologies: *PySpark, Great Expectations (GE), Iceberg, Airflow, LLM, NLP, Python, Postgres*

Custom Data Quality Rules in GE with PySpark — *Turbolab (Professional)*

- Extended Great Expectations by modifying its source code to implement field-specific custom validation rules for a streaming Spark pipeline, completing QA by the time of streaming and making end-to-end data processing 72% faster while maintaining full quality coverage.

Technologies: *PySpark, Great Expectations (GE), Python*

Pipeline Monitoring & Alert System — *Turbolab (Professional)*

- Built a Python-based pipeline monitoring system configured via a YAML file checked into the project repository; the system auto-discovers the config, monitors the pipeline, and fires real-time alerts to Google Chat on discrepancies, reducing pipeline downtime by 83% and improving availability by 67%.

Technologies: *Python, Google Chat API, YAML, Cron*

CORE COMPETENCIES

• Strategic • Decisiveness • Research • Documentation • Mentoring • Debugging • Analytical • Innovative

EDUCATION

SHM Engineering College, Kadakkal — **March 2016 - June 2020**

Computer Science and Engineering — *CGPA: 6.83*

Classic ITI, Anchal — **April 2014 - May 2015**

Computer Operation & Programming Assistant — *Marks: 72%*